

Report Period	2026-08-31 - 2026-09-07
Generated At	2026-09-07 14:33:17
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

Metric	Value	Traffic Summary	Count
Report Period	2026-08-31 - 2026-09-07	Total Flows Observed	14034
Total Threats Mitigated	47	Total Packets Analyzed	1764730051
High Priority	17	Malicious Packets	1764520273
Low Priority	30	True Negatives Passed	6265
Manual Releases	0		
Manual Blocks	0		
ML Metrics	Value	Attack Vector	Count
False Positives	0	ICMP Flood	9
FP Rate	0.00%	SYN Flood	14
IF Precision	100.00%	UDP Flood	9
IF Recall	98.33%	Uncertain	15
IF F1-Score	99.16%		
IF Accuracy	99.06%	Action	Count
		Quarantined	10
		Time Ban	10
		Blackhole	0
		Released	5

2. Performance Benchmark

2a. Isolation Forest-Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	98.33%	Share of actual attacks successfully flagged
F1-Score	99.16%	Balanced measure combining Precision and Recall
Accuracy	99.06%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	1.67%	Actual attacks that went undetected
True Positive Rate (TPR)	98.33%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 7769	FN 132
Actual: Normal	FP 0	TN 6133

2b. Random Forest-Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	4433	0	0
Act: ICMP	0	1790	0
Act: UDP	0	0	1546

2c. Response Latency

Metric	Value	Description
Detection Time	682.5 ms	Average time taken to detect an attack
Mitigation Response Time	30.0 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.90%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	10.43%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	10.43%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.24	5	3	2026-09-07 14:30:18	2026-09-07 14:32:47
10.0.0.16	3	3	2026-09-07 14:30:03	2026-09-07 14:33:09
10.0.0.19	3	3	2026-09-07 14:30:00	2026-09-07 14:33:06
10.0.0.17	2	2	2026-09-07 14:30:32	2026-09-07 14:31:39
10.0.0.18	2	2	2026-09-07 14:30:32	2026-09-07 14:31:39
10.0.0.20	2	2	2026-09-07 14:30:15	2026-09-07 14:31:34
10.0.0.21	2	2	2026-09-07 14:30:05	2026-09-07 14:31:19
10.0.0.22	2	2	2026-09-07 14:30:03	2026-09-07 14:31:23
10.0.0.23	2	2	2026-09-07 14:30:07	2026-09-07 14:31:33
10.0.0.25	2	2	2026-09-07 14:30:06	2026-09-07 14:31:29

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-07 14:30:00	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:30:03	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:30:03	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:30:05	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:30:05	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Time Ban (1m)
2026-09-07 14:30:06	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Time Ban
2026-09-07 14:30:06	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:30:08	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:30:09	10.0.0.16	DDoS	SYN Flood	91.9%	Critical	Time Ban (1m)
2026-09-07 14:30:11	10.0.0.16	DDoS	SYN Flood	91.9%	Critical	Time Ban
2026-09-07 14:30:15	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:30:16	10.0.0.21	DDoS	Uncertain	-	Low	Released
2026-09-07 14:30:17	10.0.0.25	DDoS	Uncertain	-	Low	Released
2026-09-07 14:30:18	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:30:18	10.0.0.23	DDoS	Uncertain	-	Low	Released
2026-09-07 14:30:19	10.0.0.21	DDoS	UDP Flood	78.8%	High	Time Ban (1m)
2026-09-07 14:30:19	10.0.0.21	DDoS	UDP Flood	78.8%	High	Time Ban
2026-09-07 14:30:23	10.0.0.22	DDoS	UDP Flood	80.0%	High	Time Ban (1m)
2026-09-07 14:30:25	10.0.0.22	DDoS	UDP Flood	80.0%	High	Time Ban
2026-09-07 14:30:25	10.0.0.20	DDoS	Uncertain	-	Low	Released
2026-09-07 14:30:28	10.0.0.24	DDoS	Uncertain	-	Low	Released
2026-09-07 14:30:29	10.0.0.25	DDoS	ICMP Flood	90.2%	High	Time Ban (1m)
2026-09-07 14:30:29	10.0.0.25	DDoS	ICMP Flood	90.2%	High	Time Ban
2026-09-07 14:30:32	10.0.0.23	DDoS	ICMP Flood	89.7%	High	Time Ban (1m)
2026-09-07 14:30:32	10.0.0.23	DDoS	ICMP Flood	89.7%	High	Time Ban
2026-09-07 14:30:32	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:30:32	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-07 14:30:33	10.0.0.20	DDoS	UDP Flood	74.3%	High	Time Ban (1m)
2026-09-07 14:30:33	10.0.0.20	DDoS	UDP Flood	74.3%	High	Time Ban
2026-09-07 14:30:34	10.0.0.24	DDoS	ICMP Flood	83.8%	High	Time Ban (1m)
2026-09-07 14:30:34	10.0.0.24	DDoS	ICMP Flood	83.8%	High	Time Ban
2026-09-07 14:30:38	10.0.0.17	DDoS	SYN Flood	91.0%	Critical	Time Ban (1m)
2026-09-07 14:30:38	10.0.0.18	DDoS	SYN Flood	92.1%	Critical	Time Ban (1m)
2026-09-07 14:30:39	10.0.0.18	DDoS	SYN Flood	92.1%	Critical	Time Ban
2026-09-07 14:30:39	10.0.0.17	DDoS	SYN Flood	91.0%	Critical	Time Ban
2026-09-07 14:31:05	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-07 14:31:08	10.0.0.16	DDoS	SYN Flood	91.9%	Critical	Time Ban (2m)

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-07 14:31:19	10.0.0.21	DDoS	UDP Flood	79.6%	High	Time Ban (2m)
2026-09-07 14:31:23	10.0.0.22	DDoS	UDP Flood	80.0%	High	Time Ban (2m)
2026-09-07 14:31:29	10.0.0.25	DDoS	ICMP Flood	91.7%	High	Time Ban (2m)
2026-09-07 14:31:33	10.0.0.23	DDoS	ICMP Flood	91.6%	High	Time Ban (2m)
2026-09-07 14:31:34	10.0.0.20	DDoS	UDP Flood	76.6%	High	Time Ban (2m)
2026-09-07 14:31:39	10.0.0.17	DDoS	SYN Flood	91.0%	Critical	Time Ban (2m)
2026-09-07 14:31:39	10.0.0.18	DDoS	SYN Flood	92.1%	Critical	Time Ban (2m)
2026-09-07 14:32:47	10.0.0.24	DDoS	ICMP Flood	92.4%	Critical	Time Ban (2m)
2026-09-07 14:33:06	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Time Ban (5m)
2026-09-07 14:33:09	10.0.0.16	DDoS	SYN Flood	91.9%	Critical	Time Ban (5m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.16	SYN Flood	0.6298	91.9%	Critical	Phase 2	3	3m 6s	2026-09-07 14:33:09	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6231	93.4%	Critical	Phase 2	3	3m 6s	2026-09-07 14:33:06	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6306	92.4%	Critical	Phase 2	3	6s	2026-09-07 14:32:47	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.5906	92.4%	High	Phase 2	1	1m 6s	2026-09-07 14:32:41	Ban Expired
10.0.0.24	ICMP Flood	0.6302	91.9%	Critical	Phase 2	1	6s	2026-09-07 14:31:41	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6324	91.0%	Critical	Phase 2	2	1m 7s	2026-09-07 14:31:39	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6222	92.1%	Critical	Phase 2	2	1m 7s	2026-09-07 14:31:39	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6166	91.9%	High	Phase 2	1	1m 1s	2026-09-07 14:31:35	Ban Expired
10.0.0.20	UDP Flood	0.6318	76.6%	High	Phase 2	2	1m 1s	2026-09-07 14:31:34	Escalated to Time Ban
10.0.0.23	ICMP Flood	0.6397	91.6%	High	Phase 2	2	1m 1s	2026-09-07 14:31:33	Escalated to Time Ban
10.0.0.25	ICMP Flood	0.6347	91.7%	High	Phase 2	2	1m 0s	2026-09-07 14:31:29	Escalated to Time Ban
10.0.0.22	UDP Flood	0.6240	80.0%	High	Phase 2	2	1m 20s	2026-09-07 14:31:23	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.6221	79.6%	High	Phase 2	2	1m 0s	2026-09-07 14:31:19	Escalated to Time Ban
10.0.0.16	SYN Flood	0.6407	91.9%	Critical	Phase 2	2	1m 5s	2026-09-07 14:31:08	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6142	93.4%	Critical	Phase 2	2	1m 5s	2026-09-07 14:31:05	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6443	91.0%	Critical	Phase 2	1	6s	2026-09-07 14:30:38	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6332	92.1%	Critical	Phase 2	1	6s	2026-09-07 14:30:38	Attack persisted - escalating to time ban
10.0.0.24	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-07 14:30:28	Attack Stopped
10.0.0.20	Uncertain	0.0000	-	Low	Phase 1	1	10s	2026-09-07 14:30:25	Attack Stopped
10.0.0.22	UDP Flood	0.6284	80.0%	High	Phase 2	1	20s	2026-09-07 14:30:23	Attack persisted - escalating to time ban
10.0.0.23	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-07 14:30:18	Attack Stopped
10.0.0.25	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-07 14:30:17	Attack Stopped
10.0.0.21	Uncertain	0.0000	-	Low	Phase 1	1	11s	2026-09-07 14:30:16	Attack Stopped
10.0.0.16	SYN Flood	0.6434	91.9%	Critical	Phase 2	1	6s	2026-09-07 14:30:09	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6369	93.4%	Critical	Phase 2	1	5s	2026-09-07 14:30:05	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE